Dr. NGP INSTITUTE OF TECHNOLOGY

22UAD801- PROJECT WORK
Review 2

AI-POWERED BREED IDENTIFICATION SYSTEM FOR DAIRY CATTLE AND BUFFALOES

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Objective

The objective of this project is to design and develop an **AI-powered breed identification system for dairy cattle and buffaloes** using image-based deep learning techniques. The system aims to accurately classify indigenous breeds under real-world farm conditions and **improve the quality of livestock data used for breeding, insurance, and government schemes**.

Specific Objectives:

- To automate breed identification using image-based analysis.
- To perform automated breed identification using deep learning based image classification techniques
- To provide a practical and user-friendly solution suitable for real-time usage.

Introduction

- The livestock sector plays a vital role in **India's rural economy**, supporting agricultural productivity and farmer livelihoods through dairy cattle and buffaloes.
- Accurate **breed identification** is essential for effective breeding programs, nutrition planning, disease control, insurance, and policy decisions.
- The Government of India introduced the **Bharat Pashudhan App** to digitally record breeding, health, and nutrition data of livestock.
- However, manual breed identification by field workers often leads to **misclassification,** so this project proposes an **AI-based YOLO model** to classify cattle and buffalo breeds from images.

Abstract

India is home to **50+ recognized cattle and buffalo breeds**, each with unique genetic, productivity, and disease-resistance traits. Accurate breed identification is vital for **breeding, nutrition planning, and disease control**. Currently, **manual identification by FLWs** leads to frequent misclassification, resulting in poor-quality data in national livestock databases.

This project proposes a deep learning–based breed identification system using image-based analysis, which will detect and classify breeds from captured images.

- Suggest accurate breed names in real time.
- Enhance reliability of government records and assist in effective policy-making

Problem Statement

- Manual livestock breed identification:
- Depends heavily on human expertise
- Leads to **misclassification**
- Causes **incorrect livestock database entries**
- Affects government schemes, insurance, and breeding strategies

There is **no scalable, automated, and accurate system** available for breed identification in real-time field conditions.

LITERATURE SURVEY 1

- **Image-Based Breed Recognition of Indian Cattle and Buffaloes Using YOLOv12 and Roboflow Cloud Platform (Year:2025)**
  Authors: Aashish Chauhan, Ajit Kumar Gupta, Garvit Paliwal, Gautam Singh, Harshit Singh, Dr. Umesh Dwivedi

- This study presents **BreedVision**, an AI-powered web system for automated cattle and buffalo breed recognition. The approach uses **YOLOv12 object detection** trained on **3,683 annotated images** covering 15 major Indian breeds.
- Real-time classification is achieved by integrating ReactJS frontend with a Python Flask backend and a Roboflow cloud inference engine

LITERATURE SURVEY 2

- **Development of a Lightweight Deep Learning Model for the Identification and Classification of Indigenous Cattle Breeds (Year: 2025)**
  Authors: Shruti Arya, Indu Devi, Divyanshu Singh Tomar, Man Singh, Radhika Warhade, Anshul Gautam, T. K. Mohanty

- This research focuses on **differentiating phenotypically similar indigenous cattle breeds** (e.g., Tharparkar and Hariana) using a deep learning-based segmentation and classification pipeline. This work highlights the challenge of similar visual appearance and the need for segmentation before classification in breed recognition.

LITERATURE SURVEY 3

- **Buffalo Identification in Mixed-Species Environments: A Comparative Deep Learning Approach Using ResNet50 and EfficientNetB3 (Year: 2025)**
  Authors: Nagaraj Naik (Manipal Institute of Technology), Ramu S. (Siddaganga Institute of Technology)

- This study compares the performance of **ResNet50 and EfficientNetB3** for visual buffalo identification in mixed species scenarios. By curating a balanced dataset with multiple classes (buffalo, elephant, rhino, zebra), the research evaluates how CNN architectures distinguish buffaloes from other large mammals, demonstrating the effectiveness of deeper networks in complex visual classification challenges

LITERATURE SURVEY 4

- **Ensemble Learning Algorithm for Cattle Breed Identification Using Computer Vision Techniques (Year: 2024)**
  Authors: Vijayalakshmi A, P. Shanmugavadivu, S. Vijayalakshmi, Shreyansh Padarha, Sivaranjani R

- This research proposes an **ensemble learning approach** for cattle breed identification by combining multiple computer vision techniques including CNN, YOLO object detection, Canny edge detection, k-means segmentation, and grayscale imaging. The ensemble classifier uses a voting mechanism to improve robustness against obscured features and environmental noise, suggesting that hybrid methods can address limitations of individual models.

LITERATURE SURVEY 5

- **Automatic Cattle Breed Identification Using Transfer Learning and Deep CNN (Year: 2021)**
  Authors: K. Kumar, R. Verma

- The study uses **transfer learning with deep convolutional neural networks** such as VGG16 and ResNet50 for cattle breed identification. Using pre-trained ImageNet models helps improve accuracy and reduces training time, especially when datasets are limited. Data augmentation techniques (rotation, flipping, brightness adjustment) were applied to improve model performance. The research shows that transfer learning is an effective approach for agricultural AI systems, enabling efficient and scalable breed classification.

LITERATURE SURVEY 6

- **Deep Learning-Based Animal Classification in Precision Livestock Farming (Year: 2022)**
  Authors: P. Guarnido-Lopez, J. Y. Lee

- The research reviews the use of **deep learning techniques** for applications in precision livestock farming, including breed identification, health monitoring, and behaviour analysis. It analyses architectures such as **CNNs, YOLO-based detectors, and recurrent neural networks** used in agricultural systems. The study highlights that automated livestock monitoring systems can improve farm productivity and reduce manual labour while addressing challenges like dataset variability, lighting conditions, and deployment in rural environments.

LITERATURE SURVEY 7

- **Application of Deep Convolutional Neural Networks for Fine-Grained Animal Breed Classification (Year: 2019)**
  Authors: Y. Liu, H. Zhang

- The study applies **deep convolutional neural networks (CNNs)** for fine-grained animal breed classification, where categories have very subtle visual differences. The proposed model extracts hierarchical features from images, focusing on regions such as head shape, ear position, horn curvature, and coat texture. **Attention mechanisms and deeper networks** with residual connections improve classification performance by capturing subtle variations between breeds. The research highlights that advanced CNN architectures and feature localization techniques are effective for accurate livestock breed identification systems.

Existing System

- In India, cattle and buffalo breed identification is mostly carried out through **manual visual inspection** by Field Level Workers (FLWs).
- Breed identification is based on physical characteristics such as **coat color, body structure, horn shape, and size**.
- The process depends heavily on **human expertise and experience**, making it inconsistent across regions.
- Some **AI-based research systems and online breed classifier applications** have been developed, but they are mostly limited to experimental or small-scale usage.

Existing System

- **Manual identification errors:** Breed classification depends on worker experience and training.
- **Difficulty identifying similar breeds:** Many cattle and buffalo breeds have very similar physical features.
- **Lack of AI-based validation:** No automated system to verify breed information during registration.
- **Limited AI solutions and datasets:** Existing research models have small datasets and poor real-world generalization.
- **Scalability and environmental challenges:** Variations in lighting, background, and camera angles affect accuracy, and systems are not easily scalable for nationwide use.

Proposed System

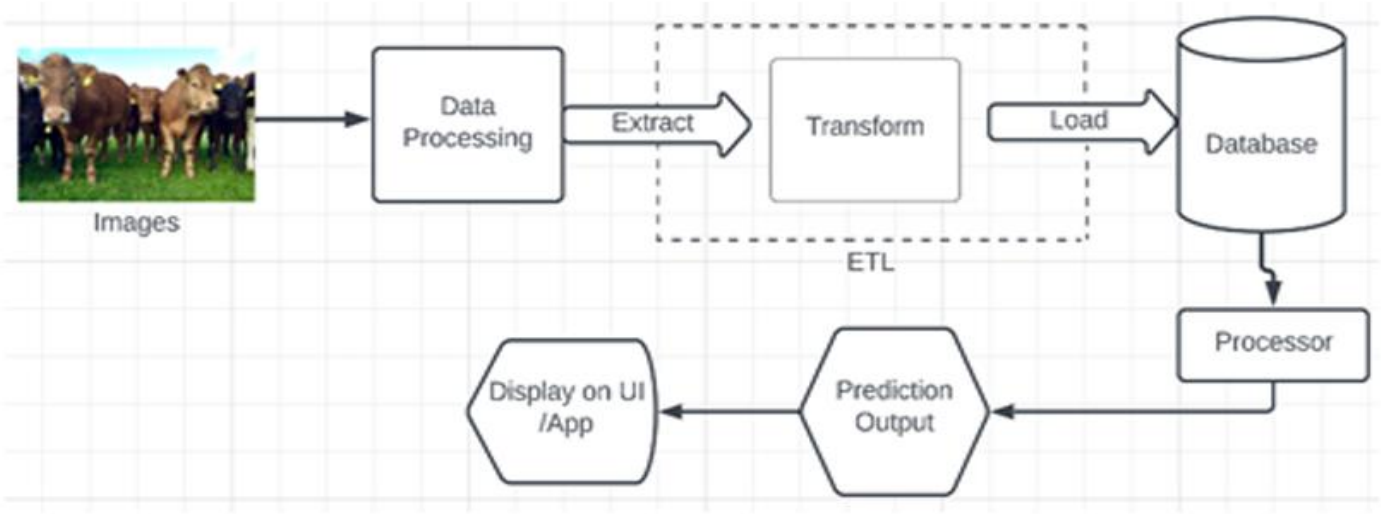
The proposed system is an **AI-powered breed identification system for dairy cattle and buffaloes** that uses a deep learning–based image classification model for breed identification

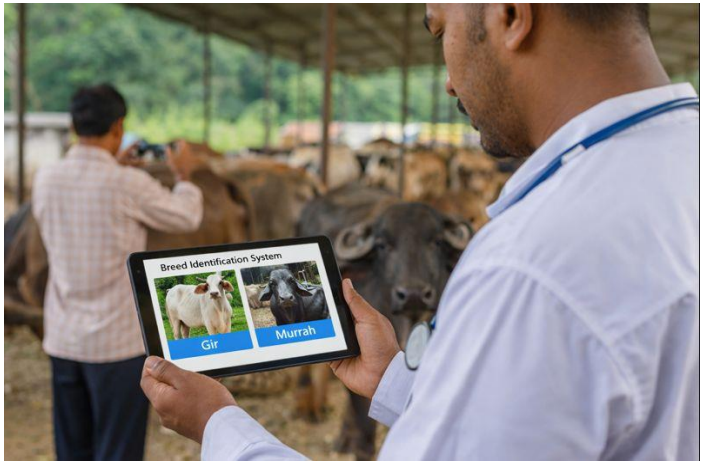
- **Key Features / Improvements:**
- Utilizes deep learning techniques for automatic feature extraction and breed classification for deep feature learning and better discrimination between visually similar breeds.
- Focuses on **classification-oriented modeling** rather than detection-first approaches.
- Robust performance under **real-world farm conditions**, including variations in lighting and background.
- Designed for **scalable deployment**, with future support for offline and edge inference.
- Can be integrated with **livestock management and government breed databases** for practical use.

Proposed System: Workflow & Architecture

- **Image Upload:** Users upload cattle or buffalo images through a web-based interface.
- **Image Preprocessing:** The system performs image resizing, normalization, and format validation to prepare the image for model processing.
- **Deep Learning Model (YOLO):** The trained model analyzes visual features such as coat pattern, horn shape, body structure, and facial characteristics to classify the breed.
- **Prediction Output:** The system displays the predicted breed along with a confidence score.
- **Future Integration:** The system can be integrated with the Bharat Pashudhan App for real-time breed validation during livestock registration.

Proposed Architecture Block diagram





Modules

1. Image Acquisition Module
2. Image Preprocessing Module
3. Feature Extraction Module (Deep Learning Model)
4. Breed Classification Module (Image-based Classifier)
5. Prediction & Result Analysis Module
6. User Interface & Deployment Module

Module Description

MODULE 1: Image Acquisition Module

Purpose:

Manages the capture or upload of cattle and buffalo images from mobile devices or cameras under real-time farm conditions.

Key Functions :

- Allows users to upload images through a **web interface**.
- Images can be captured using **mobile phones or digital cameras**.
- Supports common image formats such as **JPEG and PNG**.

Module Description

MODULE 1: Image Acquisition Module

Working:

- The module validates uploaded images before processing.
- Ensures that the image is not corrupted
- Meets minimum resolution requirements
- Simulates real farm conditions where **lighting, posture, and background may vary**.
- Provides structured input for the AI pipeline.

Module Description

MODULE 2: Image Preprocessing Module

Purpose:

Enhances and standardizes the captured images to make them suitable for deep learning model input.

Why preprocessing is needed

- Farm images may contain:
  - Uneven lighting
  - Shadows
  - Background noise
  - Resolution variations

- Preprocessing helps improve **model accuracy and reliability**.

Module Description

MODULE 2: Image Preprocessing Module

Working:

Preprocessing steps include:

- Image resizing to match YOLO model input size
- Pixel normalization
- Conversion to tensor format for
- Ensuring proper **RGB channel ordering**

Benefits:

- Standardized input data
- Reduced environmental noise
- Improved classification performance

Module Description

MODULE 3: Feature Extraction Module

Purpose:

Automatically extracts meaningful visual features from images using convolutional neural networks.

Key Features:

Automatic feature learning, extraction of texture and shape patterns, elimination of manual feature engineering

Implementation:

- Uses the **backbone network of YOLO architecture**.
- Multiple convolution layers analyze visual patterns in the image.

Module Description

MODULE 3: Feature Extraction Module

- The CNN backbone extracts:

  Low-level features

  - Edges
  - Color gradients
  - Texture patterns

  High-level features

  - Body contour
  - Horn shape
  - Facial structure
  - Coat patterns

- These features help differentiate **visually similar breeds**.

Module Description

MODULE 4: Breed Classification Module

Purpose:

Classifies cattle and buffalo breeds using a deep Residual Neural Network architecture.

Key Features:

Residual learning, deep feature refinement, high accuracy for visually similar breeds.

Function:

- Uses the **detection head of the YOLO model**.
- Performs **object localization and breed classification simultaneously**.

Module Description

MODULE 4: Breed Classification Module

The module generates:

- Bounding box around the animal
- Predicted breed label
- Confidence score

Advantages:

- Automated classification
- Reduced human error
- Real-time prediction capability

Module Description

MODULE 5 :Prediction & Result Analysis Module

Purpose:

This module processes the model output and prepares results.

Functions:

- Extract breed label and confidence score
- Verify prediction reliability using a **confidence threshold**
- Display results including:
  - Predicted breed
  - Confidence percentage
  - Annotated image with bounding box

Module Description

MODULE 6 :User Interface & Deployment Module

- Frontend developed using **HTML, CSS, and JavaScript**.
- Backend implemented using **FastAPI**.
- Handles:
  - Image upload
  - Model inference
  - Result display

Benefits:

- Easy access through web browser
- Scalable architecture
- Future integration

ALGORITHMS USED

YOLO-Based Deep Learning Model (CNN Architecture)

- Uses YOLO architecture for real-time image detection and classification.
- Extracts spatial features using convolutional neural networks (CNN).
- Detects cattle or buffalo in the image and classifies the breed category.
- Generates breed label with confidence score in a single forward pass.
- Enables fast and efficient breed identification.

ALGORITHMS USED

Multiclass Image Classification Algorithm

- Designed to classify multiple Indian cattle and buffalo breeds.
- Learns visual features from labeled bovine image datasets.
- Handles variations in:
  - Orientation
  - Lighting conditions
  - Background environment
- Provides accurate breed prediction under real farm conditions.

ALGORITHMS USED

Retrieval-Augmented Generation (RAG)

- Combines document retrieval and language generation.
- Retrieves relevant breed information from a knowledge base.
- Uses retrieved data to generate context-aware responses.
- Supports an AI-powered chatbot for breed-related queries.
- Improves accuracy and reliability of generated answers.

ALGORITHMS USED

FAISS-Based Vector Similarity Search

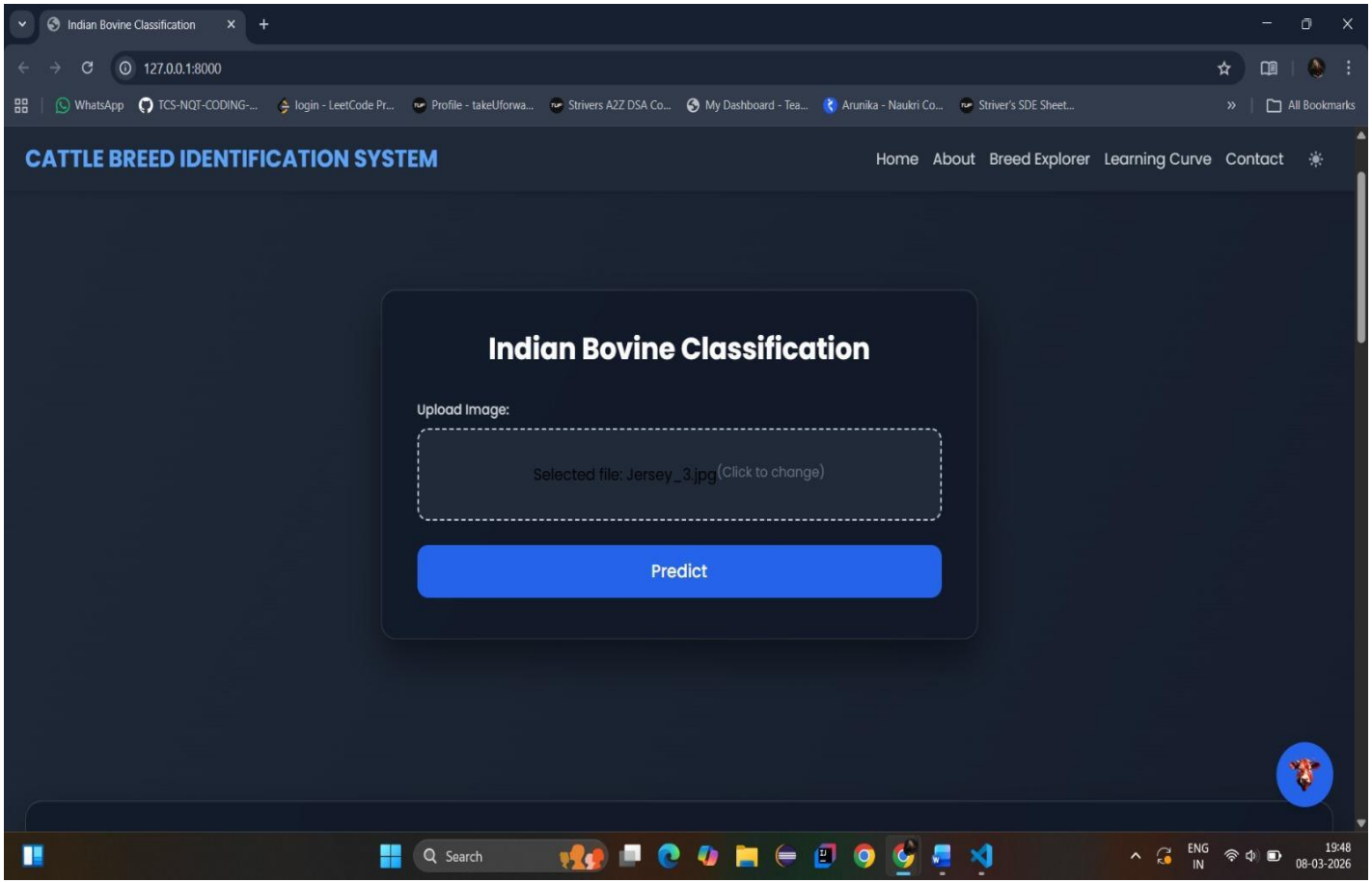
- Uses FAISS for efficient vector search.
- Indexes high-dimensional embeddings from breed documents.
- Retrieves the most relevant information quickly.
- Improves chatbot response speed and accuracy.
- Supports large-scale document retrieval systems.

ALGORITHMS USED

Large Language Model (LLM) with NLP Techniques

- Uses Large Language Models (LLMs) for natural language understanding.
- Processes user queries about Indian bovine breeds.
- Generates human-readable explanations and insights.
- Enhances user interaction and learning experience.
- Makes the system more interactive and informative.

Output Screenshot



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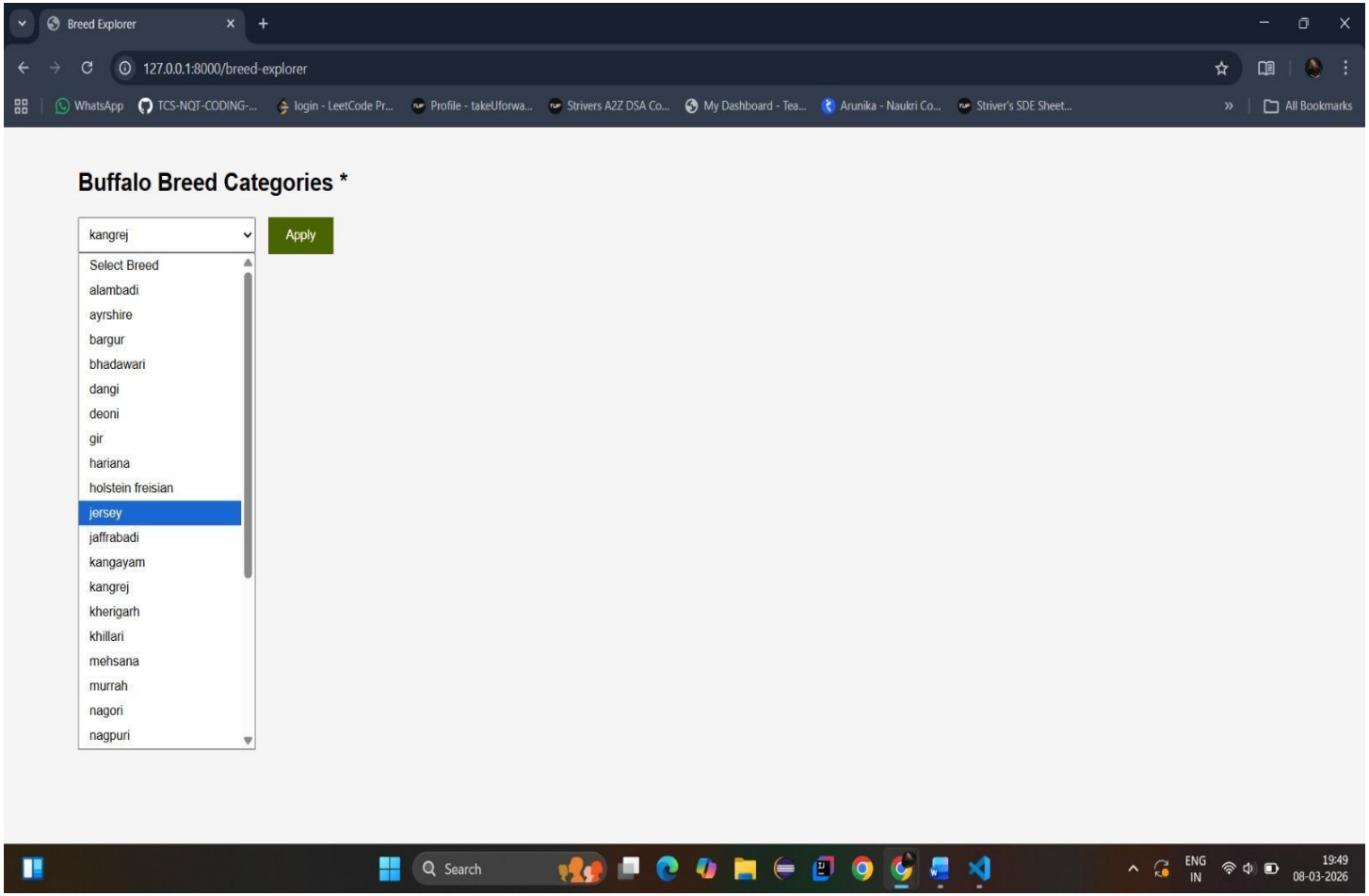

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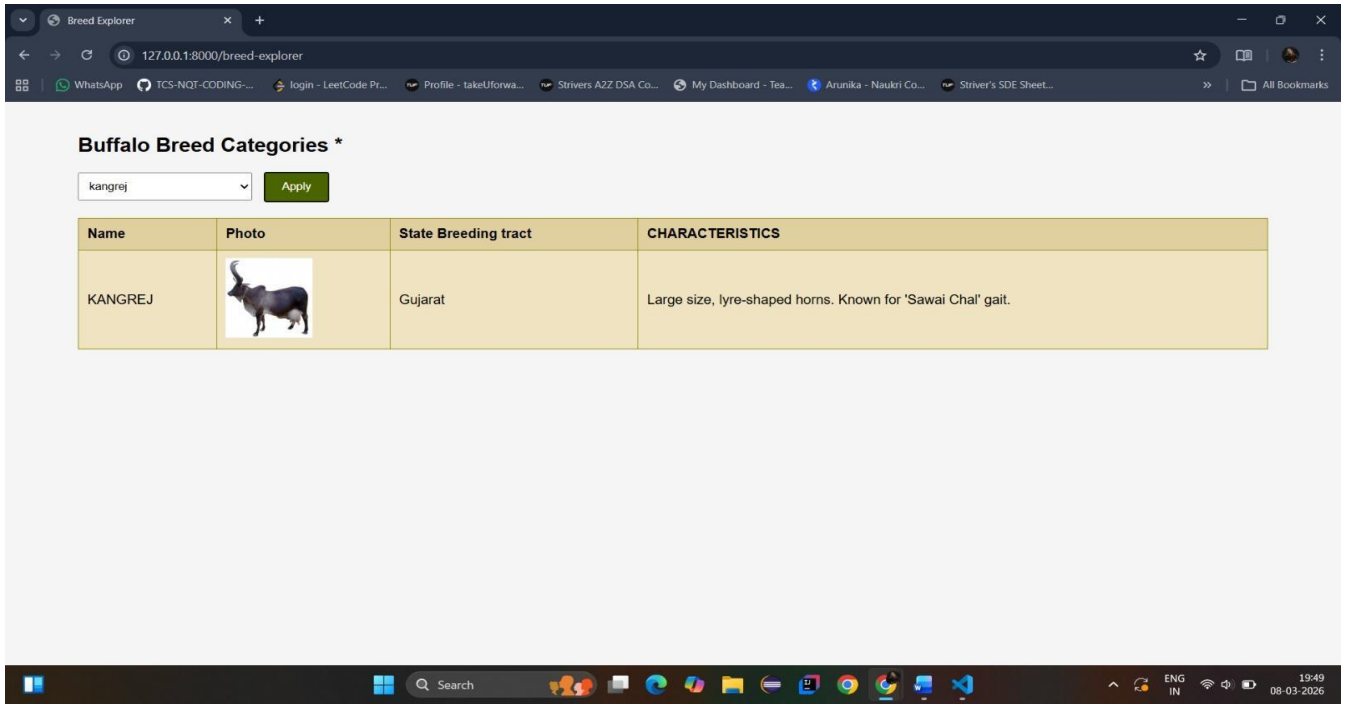

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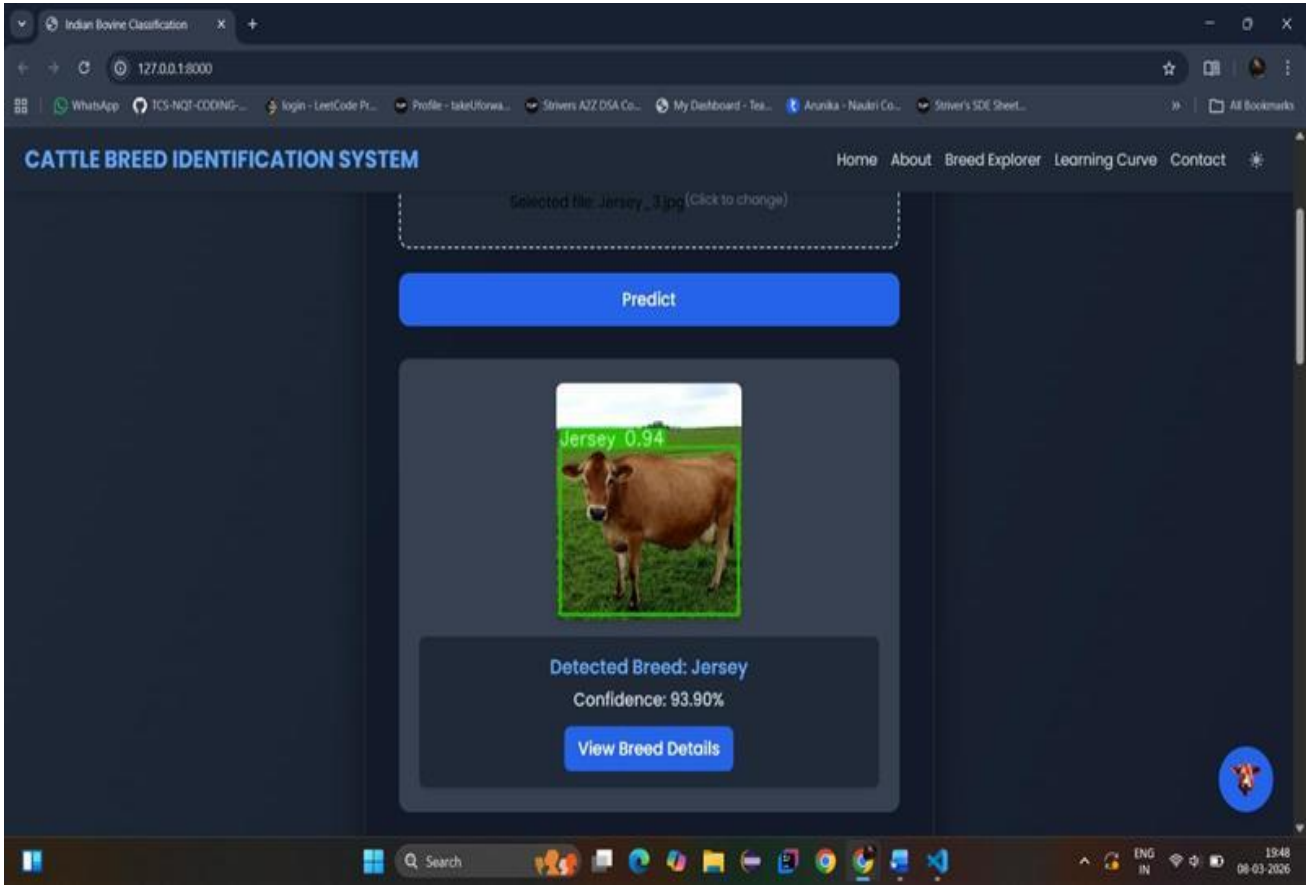

Future Enhancement

- **Integration with Bharat Pashudhan App**

  To support real-time breed validation during livestock registration by Field Level Workers.
- **Expansion of Breed Dataset**

  Include more indigenous and crossbred cattle and buffalo breeds to improve system accuracy.
- **Mobile Application Development**

  Enable field-level usage through a mobile app for easy image capture and prediction.

Conclusion

- The project demonstrates the effective use of deep learning and computer vision for automated breed identification of dairy cattle and buffaloes.
- The AI-based system reduces human dependency and improves accuracy by analyzing visual and morphological features.
- The web-based implementation enables real-time predictions with confidence scores, enhancing usability and decision-making.
- Overall, the system validates the feasibility of AI-assisted livestock management and supports the digital transformation of the agricultural sector.

References

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THANK YOU

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